Summary of the 1st International Flaky Test Workshop (FTW 2024)

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ABSTRACT

Test flakiness refers to the non-deterministic behavior of software tests. A flaky test may pass or fail without changes to the test case code or the code under test. This leads to decreased developer productivity, expensive and excessive retrying of tests, bugs masked or missed by persistent flakiness, and loss of developer trust in test suites. Despite the problem receiving increasing attention from both practitioners and researchers over the past decade, no workshop dedicated to flakiness had been established.

The 1st International Flaky Test Workshop (FTW 24), was held on April 14th 2024 at the 46th International Conference on Software Engineering (ICSE 2024). The workshop was successful in uniting developers and researchers to discuss the impact of test flakiness, propose solutions, report on the industrial state of practice, and present empirical studies.

1. INTRODUCTION

Test flakiness is a persistent problem, especially in large software projects making extensive use of automated testing and continuous integration (CI). Such projects are commonly found in industrial settings, and so connecting researchers and industrial practitioners was a primary goal of FTW 2024. This was to ensure the relevance of future research into the problem. Test flakiness has received increasing attention from both researchers and practitioners over the past several years. Unfortunately, collaboration and exchange between the two remained scarce. Therefore, the timing of FTW 2024 was opportune to combine the efforts of academia and industry to accelerate this growing area of research.

The organizing committee of FTW 2024 has an extensive background in flaky test research and software testing research in general. Owain Parry is an early-career researcher currently working on flaky tests. Martin Gruber is a 4th year Ph.D. candidate focussing on flaky tests, who recently joined the BMW Group as a Software Engineer. Tim A. D. Henderson is a Staff Software Engineer at Google working on Google's proprietary CI platform: the Test Automation Platform (TAP). Phil McMinn has been a Lecturer in the Computer Science department at the University of Sheffield since 2006. Gordon Fraser is a Professor in Computer Science at the University of Passau, Germany.

The workshop welcomed submissions on topics related to flaky tests including the costs and consequences of flaky tests, the causes of flaky tests, and techniques for the detection, mitigation, and repair of flaky tests. It was inclusive of quantitative, qualitative, and mixed-methods research, and welcomed submissions in two formats. The first was extended abstracts presenting new ideas, problems, challenges, view points, and works in progress. The second was short papers presenting technical research, experience reports, and empirical studies. In total, FTW 2024 received 7 submissions from 17 authors and accepted 6 submissions.

2. KEYNOTE

The keynote talk was delivered by Darko Marinov, titled "A Decade of Research and Teaching on Flaky Tests". The work of Darko, alongside his students and collaborators, forms an integral strand of the research literature on flaky tests. He co-authored "An Empirical Analysis of Flaky Tests", one of the earliest and most well-cited studies in the field [2]. This work introduced a range of categories for the causes of flaky tests that have been reused and adapted in many subsequent papers. Darko has also been involved in the development and scientific evaluation of automated flaky test tools, including iDFlakies [1] for detecting flaky tests and iFixFlakies [3] for repairing order-dependent flaky tests.

Darko's talk summarized the prior work of his research group on flaky tests and gave insights into how they integrated test flakiness into teaching, e.g., motivating students to fix flaky tests. The keynote concluded with a discussion of ongoing and future work. This included the automatic repair of more categories of flaky tests, approaches for detecting flaky tests sooner in the software development lifecycle (e.g., when they are introduced), and flakiness beyond software testing (i.e., the problem of nondeterminism and reproducibility in software engineering in general).

3. PAPERS

We split the paper presentations into two sessions of three. The morning session, "Mitigating Flaky Failures in CI", focussed on approaches to handle flaky test failures in a continuous integration environment. The afternoon session, "Debugging Flaky Tests in Different Domains", examined the particularities of test flakiness in specific domains, particularly artificial intelligence.

3.1 Mitigating Flaky Failures in CI

The first paper of the session was a short paper titled "Presubmit Rescue: Automatically Ignoring Flaky Test Executions" by Minh Hoang and Adrian Berding. It presents Google's latest approach to handling flaky tests, known as "Presubmit Rescue". It consists of a machine learning model that predicts the probability that a given test execution failed due to flakiness. This is as opposed to their previous approaches, which focused on predicting the probability of a given test case flaking in the future.

The second paper was the extended abstract "Regression Test History Data for Flaky Test Research" by Philipp Wendler and Stefan Winter. It explains how existing datasets used in flaky test research are mostly based on test suite reruns on an invariant code base, whereas the actual effects of flaky tests on development can only be assessed across commits on an evolving code base.

The third paper was a short paper named "Predicting the Lifetime of Flaky Tests on Chrome" by Samaneh Malmir and Peter Rigby. Their investigation of historical patterns of flaky tests in the Chrome browser revealed that 40% remain unresolved, and 33% are typically addressed within the first 10 days of introduction. The authors also trained and evaluated a machine learning model to forecast the lifespan of flaky tests.

3.2 Debugging Flaky Tests in Different Domains

The first paper of the session was the short paper "On the Impact of Hitting System Resource Limits on Test Flakiness" by Fabian Leinen, Alexander Perathoner, and Alexander Pretschner. The authors hypothesized that hitting resource limits (e.g., CPU usage) during test execution can lead to timing-related flakiness. They discovered that executions of flaky tests spend significantly more time at the CPU limit than executions of non-flaky tests.

The second paper was an extended abstract titled "Flaky Tests in the AI Domain" by Péter Attila Soha, Béla Vancsics, Tamás Gergely, and Árpád Beszédes. The authors investigate how frequently randomness is the cause of flaky tests in AI-related projects, and propose a classification scheme to promote better handling of flaky tests in such projects.

The third paper was a short paper named "Can ChatGPT Repair Non-Order-Dependent Tests?" by Yang Chen and Reyhaneh Jabbarvand. The study assesses the ability of the popular large language model ChatGPT to repair flaky tests. Their results demonstrated that ChatGPT was generally incapable of producing correct repairs for flaky tests. Finally, the authors went on to investigate the reasons why ChatGPT is ineffective at this task.

4. PANEL

The workshop concluded with a discussion panel. We selected five panel members with extensive research and industrial experience in flaky tests and software engineering more broadly. Jonathan Bell is an Assistant Professor directing research in Software Engineering and Software Systems at Northeastern University. Lionel C. Briand is a Software Engineer and Professor at the University of Ottawa and the University of Luxembourg. Mark Harman is a Research Scientist at Meta and Professor of Software Engineering at University College London. Darko Marinov (keynote speaker) is a Professor in the Department of Computer Science at the University of Illinois at Urbana-Champaign. Sigrid Eldh currently works full time leading research on Quality and Software Testing at Ericsson AB, in Stockholm, where she has worked since 1994.

Topics of discussion mainly focused on the impacts and experiences of flaky tests in industrial settings. The panel concluded with a discussion of research trends and ideas for future work.

5. CONCLUSION

FTW 2024 provided an unprecedented venue for researchers and industry practitioners to meet and discuss the problem of flaky tests. We are pleased to have been associated with ICSE 2024 and plan to propose a second edition of the workshop at ICSE 2025. The discussional and industry-focused nature of the workshop attracted positive feedback. We plan to continue and develop this format in future editions of FTW.

We would like to thank the keynote speaker, the paper authors, the discussion panel, and the program committee for making this workshop a success. We would also like to acknowledge the hard work of the ICSE 2024 organizing committee.

6. REFERENCES

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